SECTION 2

Batteries

DRY CELLS, STORAGE BATTERIES, FUEL CELLS, AND CORROSION

Standards

Electrochemistry: students explain how redox reactions convert chemical energy to electrical energy and back, and apply this to cells, batteries, corrosion, and electrolysis.

Objectives

- Which parts of a battery are the anode, the cathode, and the electrolyte?
- How are primary and secondary batteries different?
- How does a hydrogen–oxygen fuel cell work?
- Why does iron corrode, and how can corrosion be prevented?

Expectations

- Describe the structure and half-reactions of common dry cells and storage batteries.
- Classify batteries as primary or secondary and give an example of each.
- Explain the process of corrosion and two ways of preventing it.

Section 2

Batteries

MAIN IDEA

Batteries are voltaic cells that use spontaneous reactions to provide energy for a variety of purposes.

KWS

WHAT I KNOW	WHAT I WANT TO KNOW	SOURCE
		YouTube TED-Ed: How batteries work — Adam Jacobson YouTube FuseSchool: Corrosion — why metals corrode

Vocabulary

Review

- **voltaic cell** — converts chemical energy to electrical energy by a spontaneous redox reaction
- **reversible reaction** — a reaction that can take place in both the forward and reverse directions

New

- **battery** — one or more voltaic cells in a single package that generates electric current
- **dry cell** — an electrochemical cell in which the electrolyte is a moist paste
- **primary battery** — uses redox reactions that are not easily reversed
- **secondary battery** — depends on reversible redox reactions; rechargeable
- **fuel cell** — a voltaic cell in which the oxidation of a fuel produces electric energy
- **corrosion** — loss of metal from a redox reaction with substances in the environment
- **galvanization** — coating iron with a layer of zinc

Dry Cells: The Zinc–Carbon Cell

- **A battery** is one or more voltaic cells in a single package that generates electric current.
- **A dry cell** is an electrochemical cell in which the electrolyte is a moist paste.
- The paste consists of zinc chloride, manganese(IV) oxide, ammonium chloride, and a small amount of water.
- The anode is the zinc shell. The cathode is a carbon rod, but reduction occurs in the paste.

Half-reactions

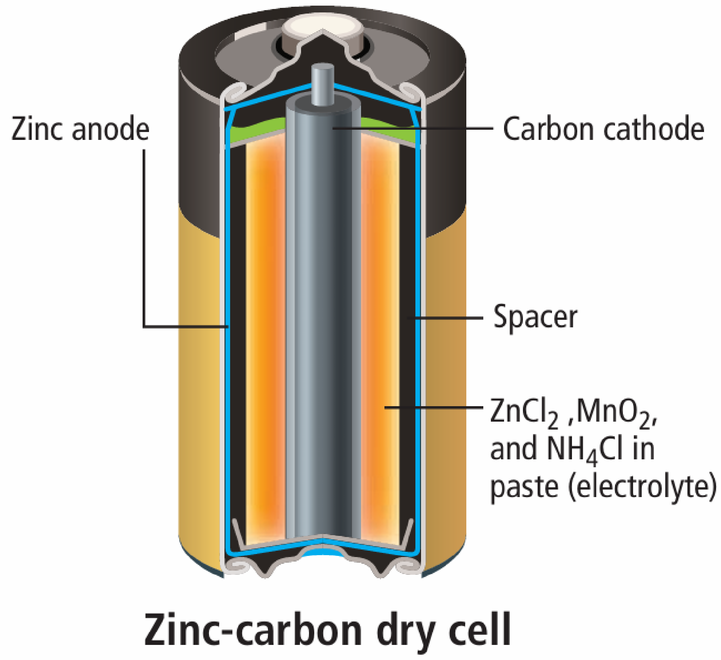
ANODE

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

CATHODE

$$2NH_4^+(aq) + 2MnO_2(s) + 2e^- \rightarrow Mn_2O_3(s) + 2NH_3(aq) + H_2O(l)$$

WATCH TED-Ed: How batteries work (4 min)

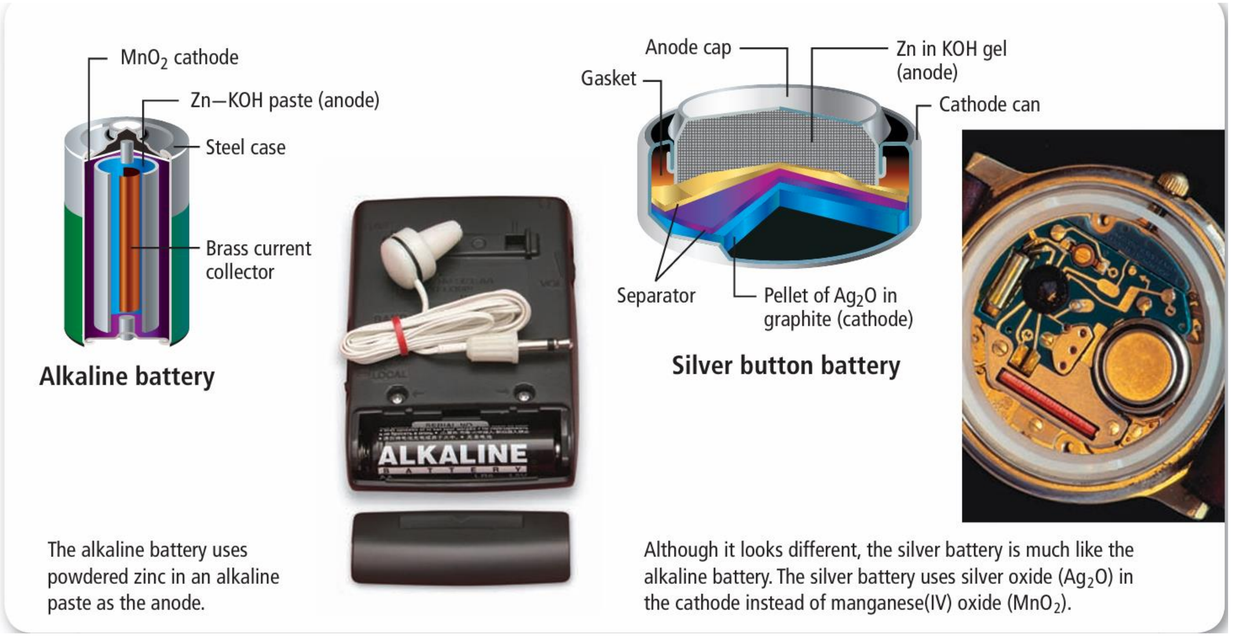


Zinc–carbon dry cell

Alkaline and Silver Batteries

- In the alkaline cell, zinc is in a powdered form and mixed with potassium hydroxide, contained in a steel case. Alkaline batteries are smaller and more useful in small devices.

- Silver batteries are similar to alkaline batteries but smaller — they use silver oxide (Ag_2O) in the cathode instead of manganese(IV) oxide.



The alkaline battery uses powdered zinc in an alkaline paste as the anode.

Although it looks different, the silver battery is much like the alkaline battery. The silver battery uses silver oxide (Ag_2O) in the cathode instead of manganese(IV) oxide (MnO_2).

Alkaline and Silver Batteries: Half-Reactions

Alkaline battery

ANODE

$$Zn(s) + 2OH^-(aq) \rightarrow ZnO(s) + H_2O(l) + 2e^-$$

CATHODE

$$MnO_2(s) + 2H_2O(l) + 2e^- \rightarrow Mn(OH)_2(s) + 2OH^-(aq)$$

Silver button battery

ANODE

$$Zn(s) + 2OH^-(aq) \rightarrow ZnO(s) + H_2O(l) + 2e^-$$

CATHODE

$$Ag_2O(s) + H_2O(l) + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$$

Same anode reaction — only the cathode material changes.

Primary and Secondary Batteries

- **Primary batteries** produce electric energy by means of redox reactions that are not easily reversed. Example: zinc–carbon, alkaline, silver.

- **Secondary batteries** depend on reversible redox reactions and are rechargeable. Example: nickel–cadmium (NiCad), lead-acid.

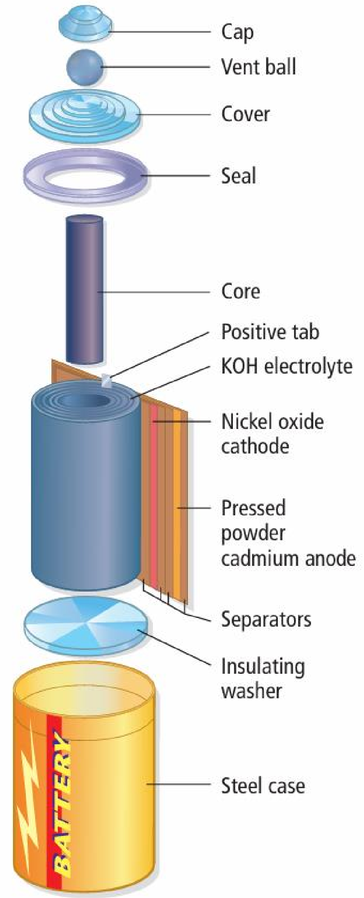
Nickel–cadmium battery (discharging)

ANODE

$$Cd(s) + 2OH^-(aq) \rightarrow Cd(OH)_2(s) + 2e^-$$

CATHODE

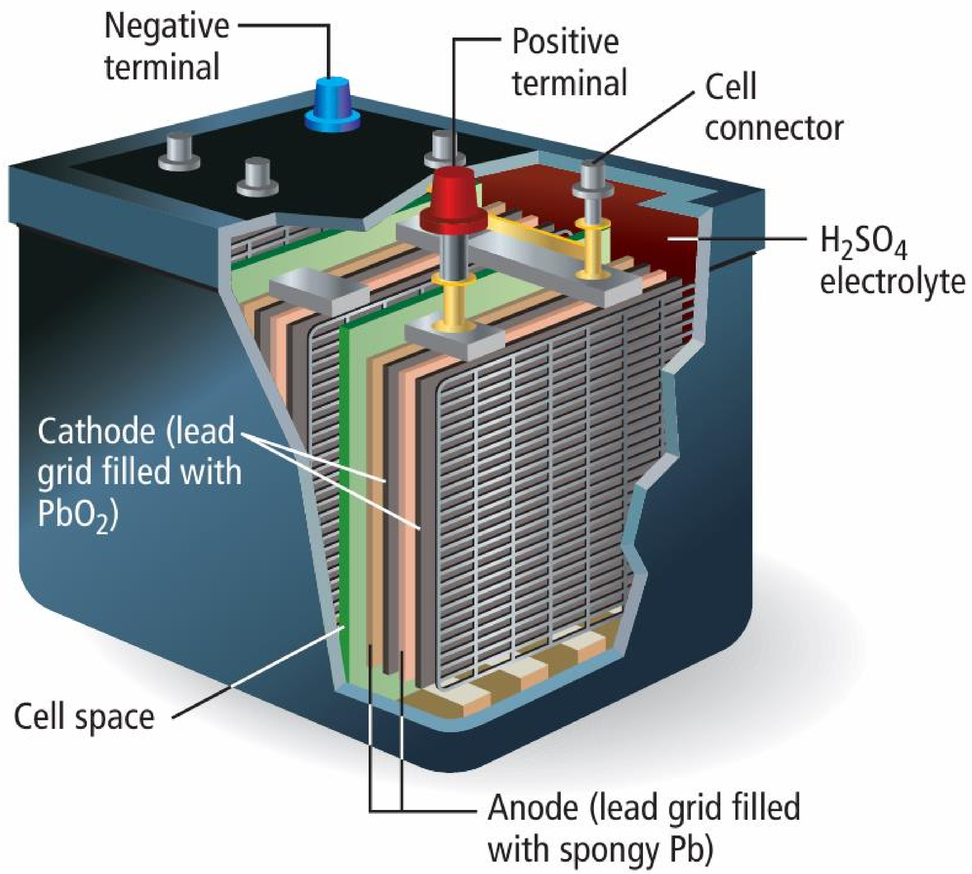
$$NiO(OH)(s) + H_2O(l) + e^- \rightarrow Ni(OH)_2(s) + OH^-(aq)$$



Inside a NiCad battery

Lead-Acid Storage Battery

- Lead-acid storage batteries are common in automobiles.
- The electrolyte solution is sulfuric acid, hence the name.
- The anode consists of grids of porous lead.
- The cathode consists of lead grids filled with lead(IV) oxide.



Lead-acid storage battery

Lead-Acid Battery: Reactions and Recharging

ANODE

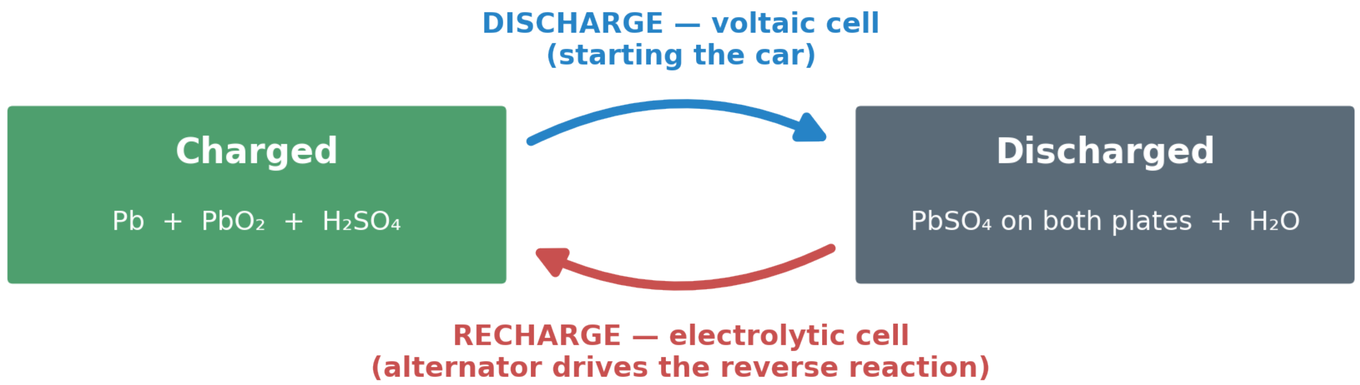
$$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$$

CATHODE

$$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$$

OVERALL

$$Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$



Lithium Batteries

- Lithium is the lightest known metal and has the lowest standard reduction potential of the metallic elements.
- Lithium batteries can be either primary or secondary.
- A lithium cell can generate about 2.3 V more than a similar cell in which zinc is oxidized.

Compare the oxidations

$E°(Zn^{2+}|Zn) = -0.762$ V

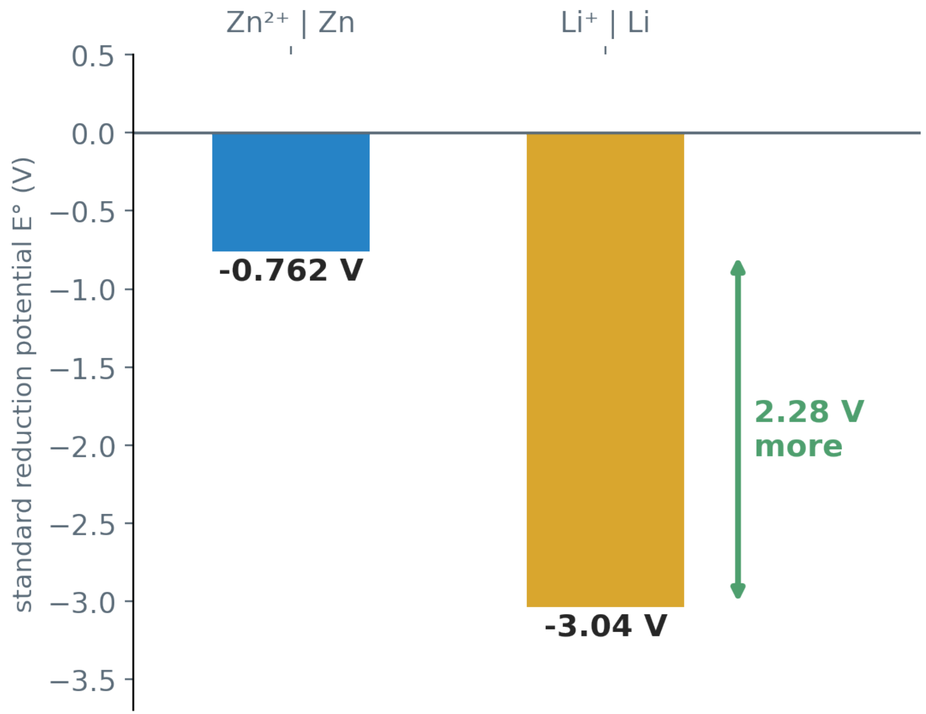
$Zn \rightarrow Zn^{2+} + 2e^-$

$E°(Li^+|Li) = -3.04$ V

$Li \rightarrow Li^+ + e^-$

DIFFERENCE

$-0.762\text{ V} - (-3.04\text{ V}) = +2.28\text{ V}$



Lithium sits far lower than zinc on the reduction-potential scale

Comparing Common Batteries

Battery	Anode	Cathode	Electrolyte	Rechargeable?
Zinc–carbon	zinc shell	carbon rod / MnO_2 paste	$ZnCl_2$, NH_4Cl paste	no (primary)
Alkaline	powdered zinc	MnO_2	KOH	no (primary)
Silver	zinc in KOH gel	Ag_2O	KOH	no (primary)
Nickel–cadmium	cadmium	NiO(OH)	KOH	yes (secondary)
Lead-acid	porous lead	PbO_2	H_2SO_4	yes (secondary)
Lithium	lithium	varies	varies	either

In every battery: oxidation at the anode, reduction at the cathode.

Fuel Cells

- **A fuel cell** is a voltaic cell in which the oxidation of a fuel is used to produce electric energy.
- Potassium hydroxide is often the electrolyte.
- When combined, the equation is the same as burning hydrogen in oxygen to form water.

Hydrogen–oxygen fuel cell (KOH electrolyte)

OXIDATION

$$2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$$

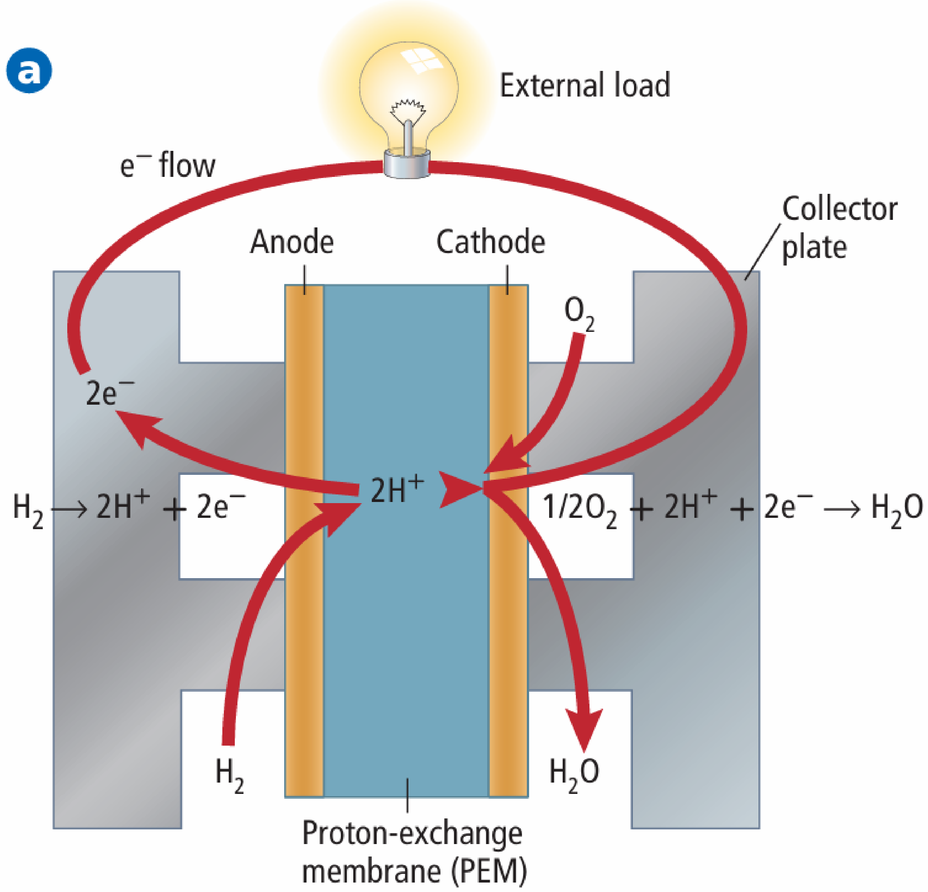
REDUCTION

$$O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$$

OVERALL

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

WATCH How Does a Fuel Cell Work?



A proton-exchange membrane (PEM) fuel cell

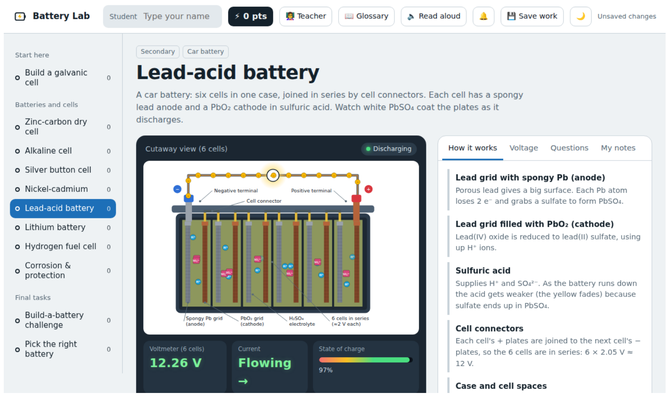
Battery Lab: Explore Each Cell

Open the Battery Lab and work through the stations for this section.

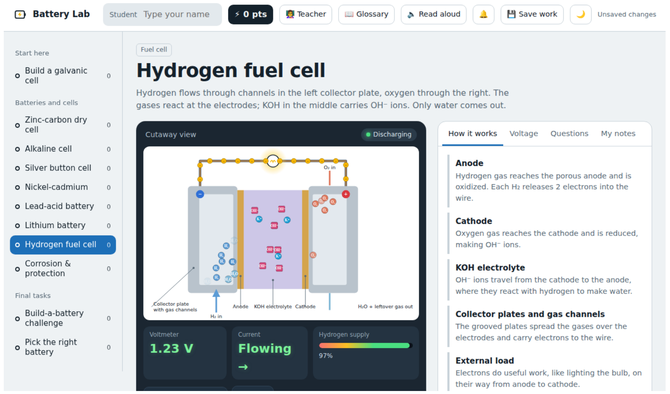
Your task

1. Visit each station: zinc–carbon, alkaline, silver, nickel–cadmium, lead-acid, lithium, fuel cell.
2. For each one, note the anode, the cathode, and the voltage.
3. Answer the questions in each station.
4. Finish with the Build-a-battery challenge and Pick the right battery.

EXPLORE Battery Lab (open the HTML file)



Lead-acid battery station

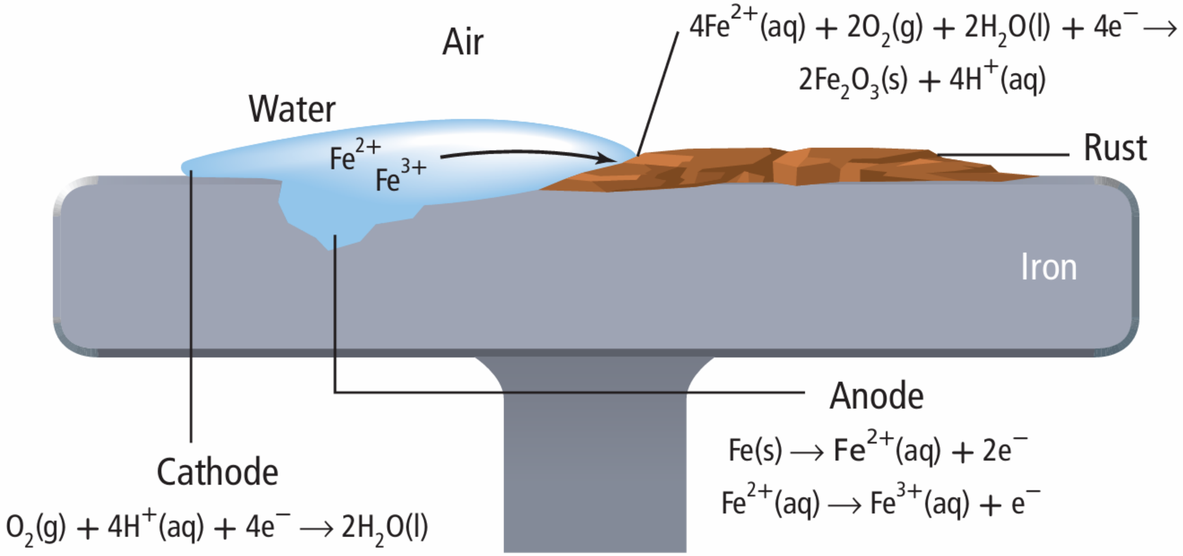


Hydrogen fuel cell station

Corrosion

- **Corrosion** is the loss of metal resulting from an oxidation-reduction reaction of the metal with substances in the environment.
- Rusting begins in a chip or pit in the iron surface, which becomes the anode. Iron(II) becomes part of the water solution.
- The cathode is usually at the edge of the water drop, where water, iron, and air come into contact.
- Rusting is slow, but salts speed the process.

WATCH FuseSchool: Corrosion



Rusting is a voltaic cell on the surface of the iron

The Chemistry of Rusting

STEP 1 — ANODE (IN THE PIT): IRON IS OXIDIZED

$$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$$

STEP 2 — CATHODE (EDGE OF THE DROP): OXYGEN IS REDUCED

$$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$$

STEP 3 — IRON(II) REACTS FURTHER TO FORM RUST

$$4Fe^{2+}(aq) + 2O_2(g) + 2H_2O(l) + 4e^- \rightarrow 2Fe_2O_3(s) + 4H^+(aq)$$

Water acts as the electrolyte — dissolved salts make it conduct better, so rusting speeds up.

Preventing Corrosion: Sacrificial Anodes

- Paint and other covers seal out moisture to prevent corrosion.
- Blocks of metal that are more easily oxidized than steel, such as magnesium, aluminum, or titanium, are often attached to the hulls of ships.
- They corrode while the iron in the hull is protected. They are called sacrificial anodes.

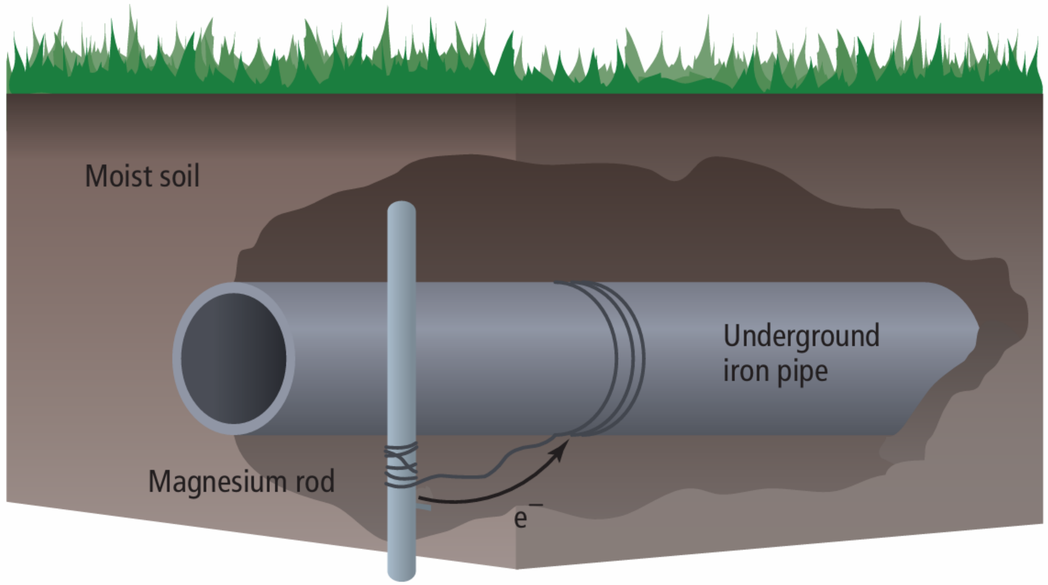
Magnesium rod protecting an iron pipe

OXIDATION

$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$

REDUCTION

$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$

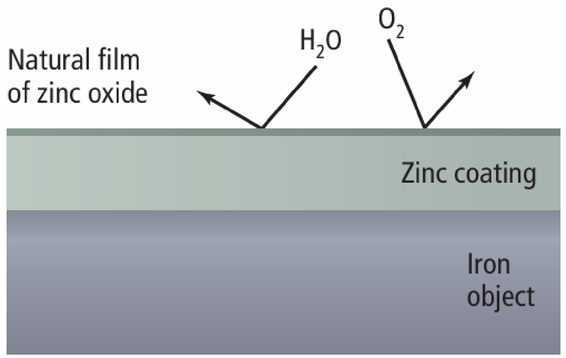


A sacrificial anode on an underground pipe

Preventing Corrosion: Galvanization

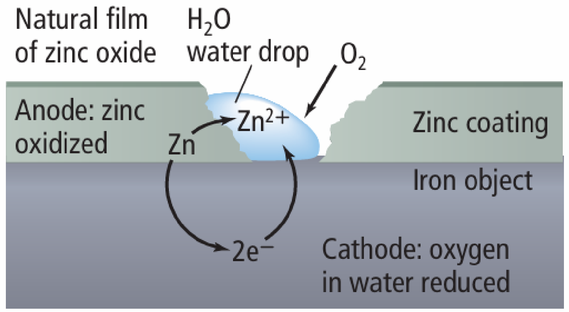
- **Galvanization** is the process of coating iron with a layer of zinc.

- Zinc oxidizes at the surface, creating a layer of metal oxide that protects from further corrosion. If the coating breaks, the zinc acts as a sacrificial anode.



Galvanized object with zinc coating intact

The zinc coating seals the iron from air and water by forming a barrier of zinc oxide that repels water and oxygen.



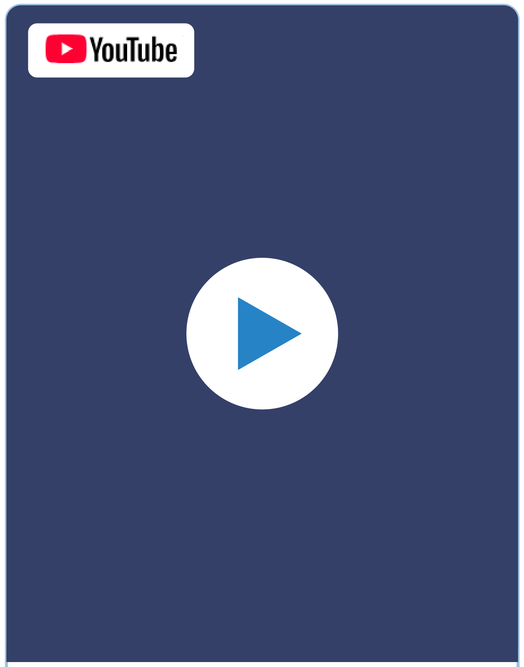
Galvanized object with zinc coating broken

If the zinc coating breaks, the zinc acts as a sacrificial anode. The zinc coating, rather than the iron object, is oxidized.

WATCH Cognito: What is Corrosion and How to Stop it

Watch & Explore: Batteries and Corrosion

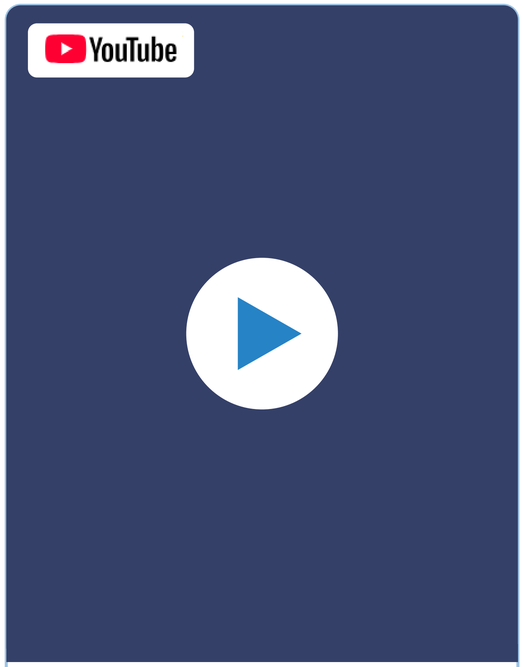
Click a card to open it.



How batteries work

TED-Ed • Adam Jacobson • 4 min

Why batteries store charge — and why they die.



How Does a Fuel Cell Work?

YouTube

Hydrogen and oxygen making electricity and water.



Corrosion

FuseSchool

Why metals corrode.



Battery Lab

Class lab (HTML file)

Explore every battery in this section, then take the challenges.

Checkpoint: Batteries

1. In a zinc–carbon dry cell, which part is the anode?

2. What is the difference between a primary battery and a secondary battery? Give one example of each.

3. Why is sulfuric acid needed in a lead-acid battery?

4. What is the only product of a hydrogen–oxygen fuel cell?

5. Why does a magnesium block protect a steel ship hull?

Quick check — discuss with a partner, then confirm with your teacher.

Lesson Summary

Key Takeaways

- Primary batteries can be used only once; secondary batteries can be recharged.
- When a battery is recharged, electric energy supplied to the battery reverses the direction of its spontaneous reaction.
- Fuel cells are batteries in which the substance oxidized is a fuel from an external source.
- Methods of preventing corrosion are painting, coating with another metal, or using a sacrificial anode.

Vocabulary Recap

- **battery** — one or more voltaic cells in one package
- **dry cell** — electrolyte is a moist paste
- **primary / secondary** — single-use / rechargeable
- **fuel cell** — oxidation of a fuel produces electric energy
- **corrosion** — loss of metal by a redox reaction
- **galvanization** — coating iron with zinc

Evaluation: Batteries

Complete independently. Review the summary if you get stuck, then check your answers with your teacher.

1. Identify the anode, the cathode, and the electrolyte in a zinc–carbon dry cell.

2. Explain how primary and secondary batteries differ. Give an example of each type.

3. Explain why a fuel cell does not run down like other batteries.

4. Describe what happens at the anode and the cathode when iron rusts.

5. Explain why lithium is a good choice for the anode of a battery.

6. **Challenge:** A steel pipe can be protected with a block of magnesium or by galvanizing it with zinc. Compare the two methods: what do they have in common, and when would you choose each one?

Self-Evaluation Rubric

4 – Confident mastery • 3 – Solid grasp, minor gaps • 2 – Needs review • 1 – Revisit the lesson